25901745 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCPEG100 - DRILLING, TRANSPORTATION ANDREFINING TECHNIQUES

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple choice questions

Answer all questions

Each question carries **1** marks

1. How can water circulation contribute to scaling in pipelines? [K] / [CO1]

   a). By dissolving all minerals

   b). By decreasing the salinity of water

   c). By depositing calcium carbonate and sulfate compounds

   d). By reducing pressure in pipelines

2. The presence of sulfate-reducing bacteria (SRB) in oil field water leads to : [U] / [CO1]

   a). Increased oil viscosity b). Improved water clarity

   c). Reduced corrosion risk d). Hydrogen sulfide production

3. What is the main purpose of secondary oil recovery methods like gas flooding and water flooding? [U] / [CO1]

a). To decrease the viscosity of the oil for faster extraction.

b). To extract all the oil from the reservoir in one go.

c). To change the composition of the oil to make it more usable

d). To increase the pressure in the reservoir and extract more oil after primary recovery.

4. What is an advantage of GTL fuels over traditional petroleum fuels? [U] / [CO1]

a). Higher sulfur content

b). Limited availability of raw materials

c). Lower emissions and cleaner burning

d). More impurities

5. A company wants to produce a wide range of refined products, including high-value petrochemicals and specialized fuels. Which type of refinery should they choose? [U] / [CO3]

a). Simple refinery b). Complex refinery

c). Integrated refinery d). Crude oil storage facility

6. Which distillation method would you recommend for separating a mixture of compounds with boiling points differing by less than 5°C? [E] / [CO4]

a). Fractional distillation b). Vacuum distillation

c). Simple distillation d). Steam distillation

7. Identify the heat exchanger type that uses concentric tube construction: [K] / [CO5]

a). Plate heat exchanger

b). Shell and tube heat exchanger

c). Double-pipe heat exchanger

d). Spiral heat exchanger

8. High concentrations of calcium and sulfate in brines can lead to the formation of : [U] / [CO1]

a). Sodium chloride crystals b). Barium sulfate scaling

c). Gypsum scaling d). Iron oxide deposits

9. Specific gravity is the ratio of the density of a substance to the density of: [K] / [CO6]

a). Air b). Mercury

c). Water d). Oil

10. The standard method used to determine the cloud point: [U] / [CO6]

a). ASTM D97 b). ASTM D86

c). ASTM D2500 d). ASTM D445

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **3** marks

11. Explain the dissolved gas drive in primary oil recovery. [U] / [CO1]

12. List out the safety consideration of nitrogen blanketing. [K] / [CO2]

13. Describe the composition of fuel oil and its applications. [K] / [CO4]

14. What is Natural Rubber Modified Bitumen (NRMB)? [K] / [CO4]

15. Discuss the main functions of drilling mud in oil and gas drilling. [U] / [CO1]

16. Explain the principle of the Abel flash point test. [U] / [CO6]

[3x4 = 12]

Part C

Short Essay Questions

Answer any **4** questions

Each question carries **7** marks

17. Discuss the main safety concerns on oil rigs, and how can they be reduced. [U] / [CO1]

18. Compare and contrast the environmental risks of transporting oil and gas by pipeline versus by tanker (ship). [U] / [CO2]

19. Explain the composition, functions and properties of Light Diesel Oil and Lubricating oil. [U] / [CO4]

20. Discuss the differences between Aromatic Naphtha (HAN) and Paraffinic Naphtha (LAN) in terms of composition and applications. [U] / [CO4]

21. Describe the working principle of atmospheric and vacuum distillation. Compare their fraction seperated and applications. [U] / [CO6]

22. Explain the procedure and significance of the penetration test for bitumen. [U] / [CO6]

[7x4 = 28]